
9

Medians and Order Statistics

The i th *order statistic* of a set of n elements is the i th smallest element. For example, the *minimum* of a set of elements is the first order statistic ($i = 1$), and the *maximum* is the n th order statistic ($i = n$). A *median*, informally, is the “halfway point” of the set. When n is odd, the median is unique, occurring at $i = (n + 1)/2$. When n is even, there are two medians, the *lower median* occurring at $i = n/2$ and the *upper median* occurring at $i = n/2 + 1$. Thus, regardless of the parity of n , medians occur at $i = \lfloor (n + 1)/2 \rfloor$ and $i = \lceil (n + 1)/2 \rceil$. For simplicity in this text, however, we consistently use the phrase “the median” to refer to the lower median.

This chapter addresses the problem of selecting the i th order statistic from a set of n distinct numbers. We assume for convenience that the set contains distinct numbers, although virtually everything that we do extends to the situation in which a set contains repeated values. We formally specify the *selection problem* as follows:

Input: A set A of n distinct numbers¹ and an integer i , with $1 \leq i \leq n$.

Output: The element $x \in A$ that is larger than exactly $i - 1$ other elements of A .

We can solve the selection problem in $O(n \lg n)$ time simply by sorting the numbers using heapsort or merge sort and then outputting the i th element in the sorted array. This chapter presents asymptotically faster algorithms.